

H.A.V.E | 00 Whole-system overview

Accepted ERD v1.0 • 27 September 2026 • 21 application tables • Exact fields and crow's-foot FK relationships are on the five detail tabs.
PK / FK shown inline. ? = nullable; other fields required. All timestamps UTC; money is decimal EGP.

ACCOUNTS & OWNER APPROVAL

users
customers
doctors
owners
refresh_tokens

CUSTOMER / DOCTOR / OWNER identities.
Phone + email unique within account type.

CATALOGUE & INVENTORY

categories
products
product_batches
terminals
terminal_slots
terminal_stock

Stock belongs to a terminal slot and batch.
Device credential hash is on terminals.

PRESCRIPTIONS

prescriptions
prescription_products

Customer link can be NULL before registration.
Assigned phone fixed; account names / DOB win.

ORDERS & FULFILMENT

orders
order_products
stock_reservations

Guests have no customer FK.
Confirmed line quantities support partial orders.

MOCK INSURANCE

customer_insurances
insurance_products

Covered products + expiry + monthly uses.
Coverage configurable; default 75%.

PAYMENT & RECEIPTS

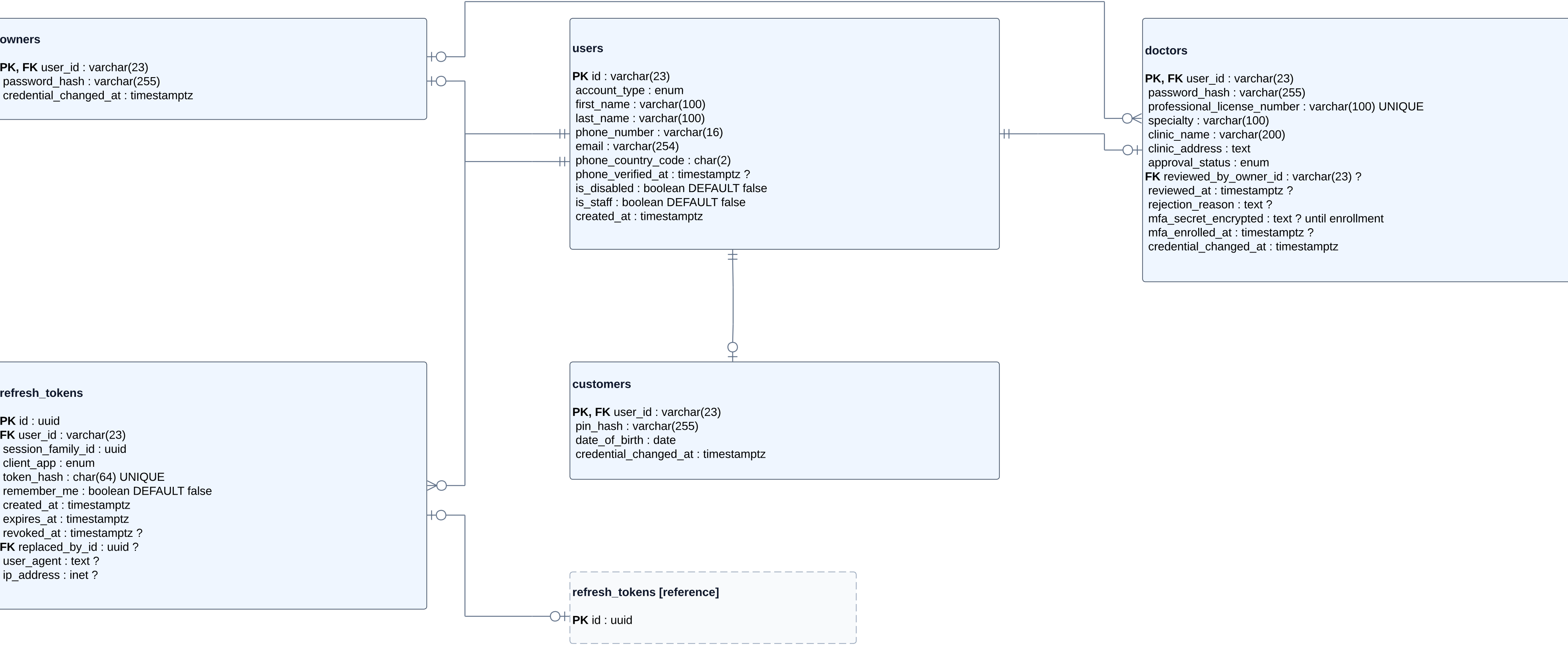
saved_payment_methods
payment_transactions
receipts

Mock digital payments; no physical card reader.
Opaque references only; no PAN / CVV.

Session limits: Hub / Advisor access 15 min, refresh 1 day or 7 days with Remember Me. Terminal: 70-second inactivity / 10-minute maximum. Owner: Django native session, 10-minute inactivity / 2-hour maximum.
Not tables: guidance rules, OTP challenges, generic audit/outbox events, approval codes, separate terminal credentials, dispense attempts, insurance plans/claims.
This is the accepted future design. Existing backend code and historical diagrams are not migrated or replaced.

H.A.V.E | 01 Accounts & doctor approval

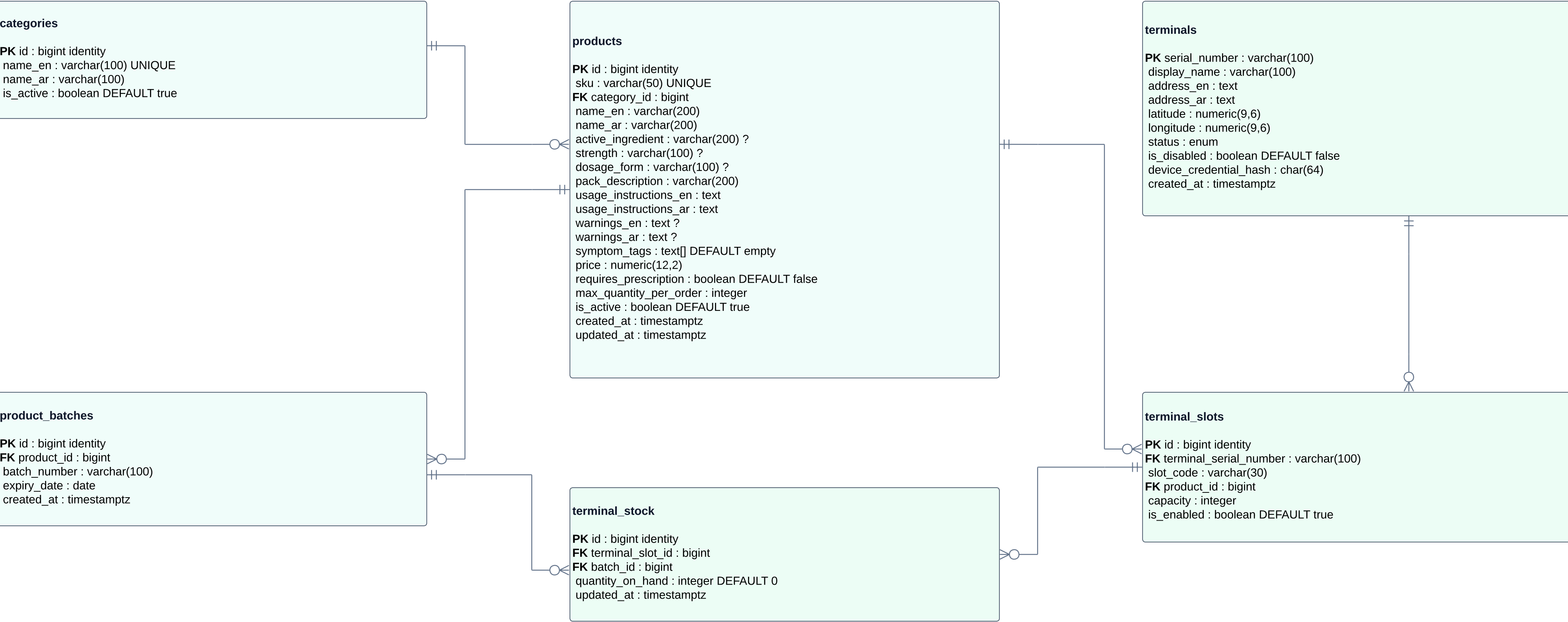
One shared identity; exactly one matching subtype. Owner is provisioned privately. References repeat existing tables rather than introducing new tables.
PK / FK shown inline. ? = nullable; other fields required. All timestamps UTC; money is decimal EGP.



account_type = CUSTOMER / DOCTOR / OWNER. approval_status = PENDING / APPROVED / REJECTED. client_app = HUB / ADVISOR.
Unique (account_type, phone_number) and (account_type, lower(email)). Every subtype requires its own hash. IDs: HAV-C887-000000000001 (D for doctor; O for owner), sequence-generated and immutable.
Refresh tokens rotate atomically; expiry is fixed at original login. Keep revoked/replaced evidence until expiry. Temporary challenges and Terminal sessions stay in Redis. Django admin uses native Django session records, omitted as framework infrastructure.

H.A.V.E | 02 Catalogue & terminal inventory

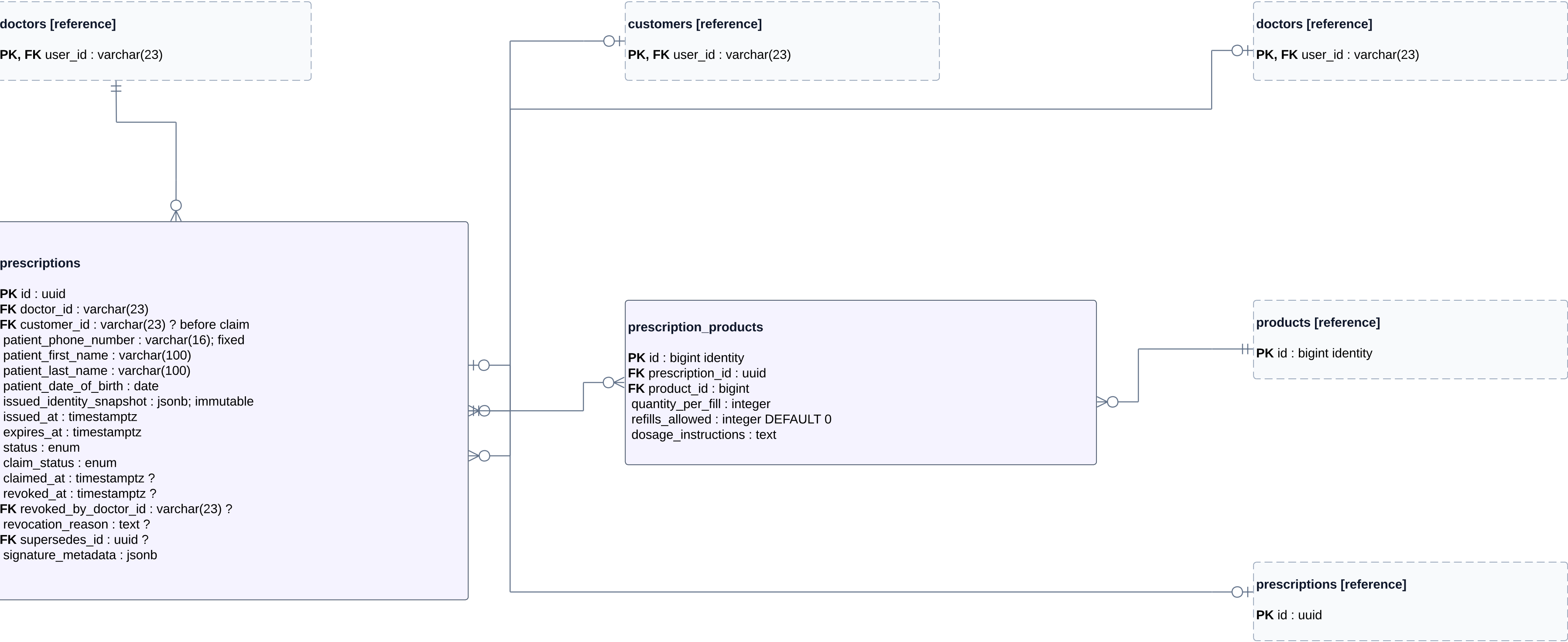
Products are machine-dispensable. Stock allocation is per slot and batch, not a global quantity on products.
PK / FK shown inline. ? = nullable; other fields required. All timestamps UTC; money is decimal EGP.



terminal status = ONLINE / OFFLINE / MAINTENANCE / OUT_OF_SERVICE. is_disabled defaults false. Heartbeat freshness lives in Redis.
Unique (product, batch_number), (terminal, slot_code), (slot, batch). Batch product must match slot product. Quantities are packs; stock cannot be negative or exceed slot capacity.
Guidance safety/ranking rules are versioned files; product symptom_tags are metadata. Reservations and order lines appear on the Orders tab. Stock decreases only after confirmed dispensing.

H.A.V.E | 03 Prescriptions & patient assignment

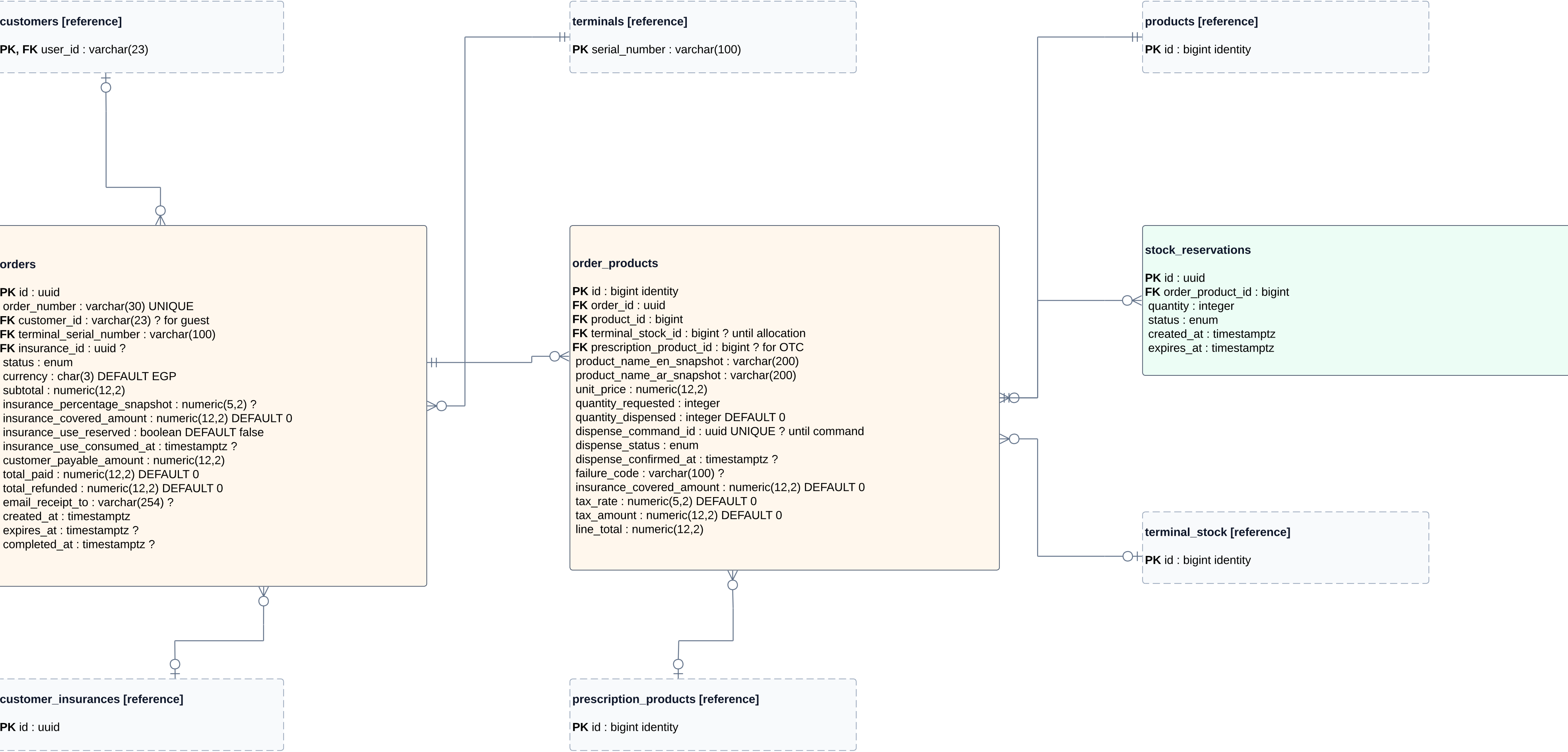
A doctor can prescribe before a customer account exists. The assigned phone is permanent; account names / DOB take display priority after verified claim.
PK / FK shown inline. ? = nullable; other fields required. All timestamps UTC; money is decimal EGP.



status = ISSUED / REVOKED / SUPERSEDED. claim_status = UNCLAIMED / CLAIMED / DISPUTED. Expiry and remaining quantities are derived.
After verified exact-phone claim, overwrite working first_name / last_name / DOB with customer account data EVEN WHEN THE MANUAL NAME DIFFERS. Never overwrite patient_phone_number from account edits. The issued snapshot is historical, not the displayed identity.
Unique (prescription_id, product_id). Total pack allowance = quantity_per_fill × (refills_allowed + 1). Confirmed order-line quantities determine usage. Existing-account lookup searches customers only.

H.A.V.E | 04 Orders, allocation & fulfilment

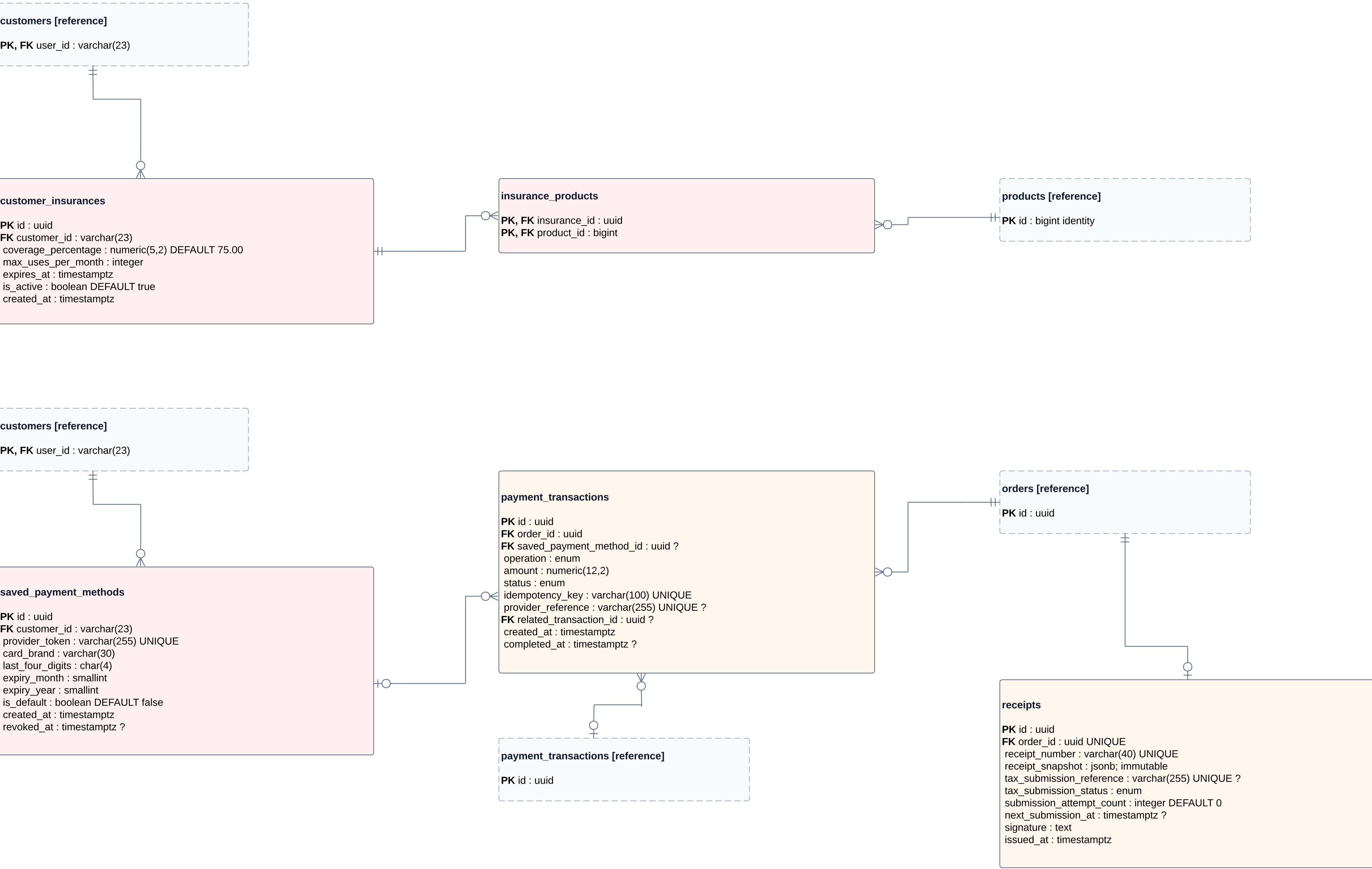
Guest purchases are supported. Each allocated order line references one stock row / batch; split lines when sourcing the same product from multiple batches.
PK / FK shown inline. ? = nullable; other fields required. All timestamps UTC; money is decimal EGP.



Order status = PENDING / DISPENSING / CANCELLED / COMPLETED / REVIEW_REQUIRED. dispense_status = PENDING / SUCCEEDED / FAILED / UNKNOWN.
Reservation status = ACTIVE / CONSUMED / RELEASED / EXPIRED. Stock target is obtained through the order line. No separate dispense-attempts table.
Confirmed quantities, stable command IDs, snapshots, and payment outcomes—not extra status transitions—describe partial completion. UNKNOWN outcomes require reconciliation, never a blind retry. Lock stock, prescription allowance, and insurance usage during reservation/confirmation.

H.A.V.E | 05 Mock insurance, payment & receipts

Local configurable insurance and mocked digital payments. Repeated reference cards point to the same customer/order table.
PK / FK shown inline. ? = nullable; other fields required. All timestamps UTC; money is decimal EGP.



Insurance default coverage 75%, configurable 0–100%. Covered-product links have a composite PK. Monthly usage and active usage reservations are counted through orders under a transaction lock (Africa/Cairo calendar month).
Payment operation = AUTHORIZE / CAPTURE / VOID / REFUND. Payment status = PENDING / SUCCEEDED / FAILED / UNKNOWN. ETA status = PENDING / ACCEPTED / REJECTED. One active default saved method per customer; no raw card/CVV data.
Receipt order FK is unique. Pending receipt/submission records support retries without a generic outbox. Financial and issued snapshots are preserved; security logs contain no secrets.